



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to D.V.S FUEL STATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

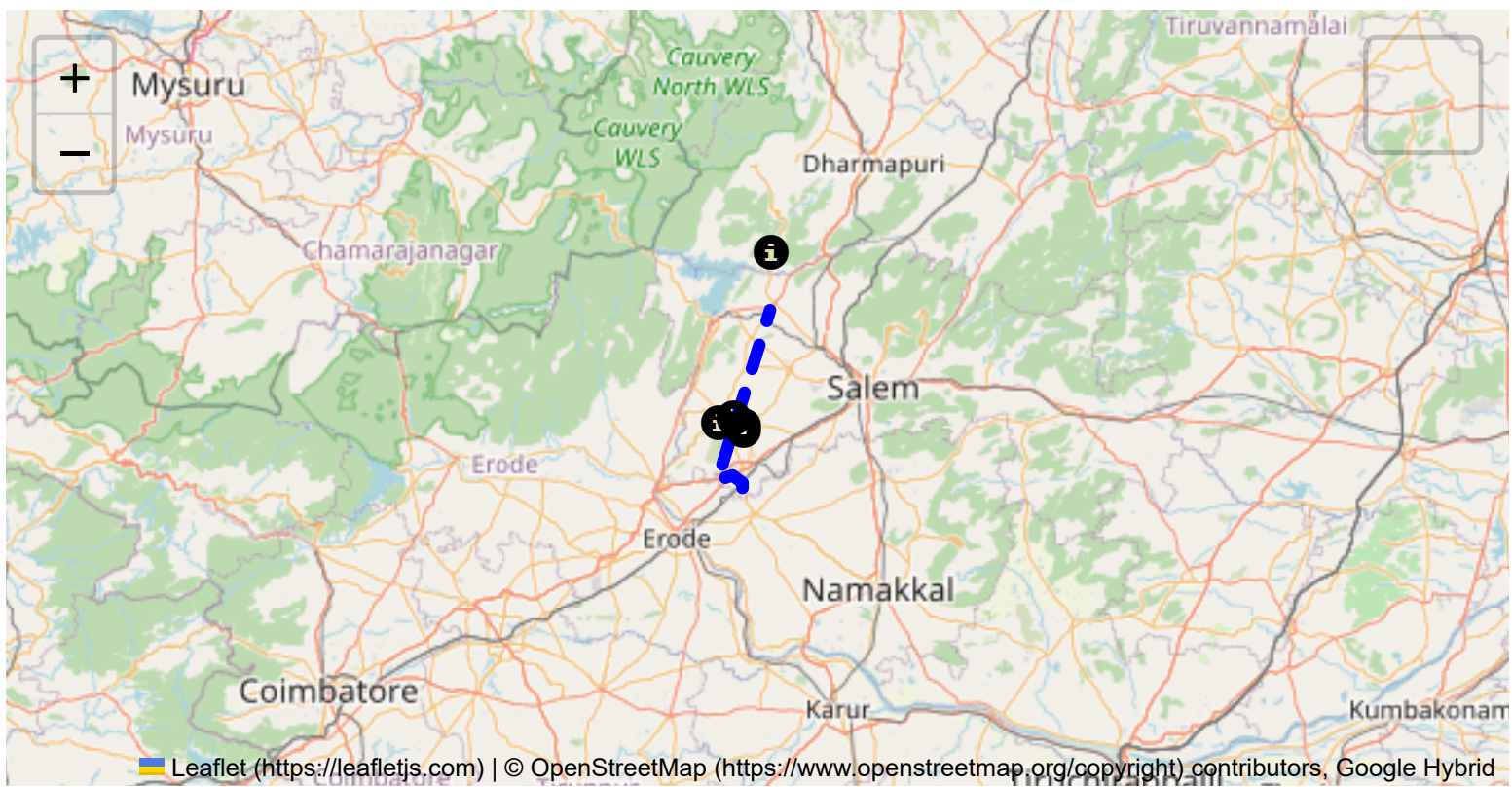
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 58.38 km
Estimated Duration: 1.4 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.81891, 77.93381)



Welcome to the Journey Risk Management Study

Route Map Overview

The route spans approximately 58.38 kilometers in Tamil Nadu, India, originating at Sangagiri and terminating at M. Kalipatti. This route primarily utilizes state highways and regional roads, passing through notable points like Sankari RS - Erode Privu Road and Bhavani Main Road, traversing a mix of urban and semi-urban areas.

Typical Weather Conditions

- **Seasonal Patterns:** The region experiences a tropical climate. Summers (March to June) can be hot, whereas the monsoon (July to September) might bring heavy rains.
- **Weather-Related Hazards:** During the monsoon, roads might be slippery, with potential roadway flooding. Extreme heat in summers could lead to road fatigue or vehicle overheating.

Traffic Patterns

- **Peak Hours:** Morning (8-10 AM) and evening (5-7 PM) times typically see increased traffic due to daily commuters.
- **Congestion-Prone Areas:** Areas around towns like Sankari and Padaiveedu might experience traffic jams, especially near commercial and market areas.

Road Quality and Infrastructure

- **General Conditions:** The roads are mostly two-lane highways, with varying quality. Some sections might have potholes, especially after the rainy season.
- **Infrastructure Notes:** Limited street lighting in rural sections, but adequate signage throughout the route.

Alternative Routes for Emergencies

- An alternative route could include bypasses through adjacent villages to avoid congested town centers, though such routes might be narrow and less maintained.
- Utilizing the National Highway 544 can provide a faster alternative if congestion is severe in local segments.

Local Regulations on Hazardous Material Transport

- **Permits:** Required for transporting hazardous materials; ensure all documentation is up-to-date.
- **Timings:** Some areas might restrict hazardous material transport during peak hours in populous regions.

Historical Incidents

- **Previous Incidents:** No major incidents in recent years involving heavy vehicles on this particular route, but caution is advised near intersections, where minor collisions have occurred.

Environmental Considerations

- **Sensitive Areas:** Proximity to agricultural lands and small water bodies; precaution to prevent contamination from potential spills.
- **Conservation Zones:** None directly along the route, but awareness and care are crucial.

Communication Coverage

- **General Assessment:** Coverage by major telecom operators is good, but some rural stretches might experience brief dead zones.
- **Recommendations:** Carry backup communication methods, such as satellite phones, for emergencies.

Emergency Response Times

- **Rural Areas:** Longer response times due to distance from urban centers, roughly estimated at 30-45 minutes.
- **Urban Areas:** Faster response, approximately 15-30 minutes, depending on specific locations and traffic conditions.

Overall Risk Assessment

- **Projecting Risk Levels:** The route is broadly safe but can present challenges due to variable road quality, weather conditions, and traffic congestion.
- **Critical Recommendations:** Regular road condition assessments post-monsoon, adherence to driving safety norms, well-maintained communication tools, and periodic driver training updates, especially concerning hazardous material regulations. Further consideration for scheduling transport during non-peak hours to mitigate traffic risks is advisable.

By addressing these insights and recommendations, a truck driver can navigate this route more safely, ensuring the protection of both the transported goods and the surrounding environment.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60348, 77.83967	15 KM/Hr
2	Turn	Medium	11.43822, 77.87348	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44900, 77.87406	30 KM/Hr
6	Turn	Medium	11.45357, 77.85789	30 KM/Hr
7	Turn	High	11.45352, 77.85698	15 KM/Hr
8	Turn	Medium	11.45847, 77.85140	30 KM/Hr
9	Turn	Medium	11.46303, 77.84945	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.46314, 77.84928	30 KM/Hr
11	Turn	High	11.45542, 77.81725	15 KM/Hr
12	Turn	High	11.45634, 77.81673	15 KM/Hr
13	Blind Spot	Blind Spot	11.49390, 77.86748	10 KM/Hr
14	Turn	Medium	11.49404, 77.86724	30 KM/Hr
15	Turn	Medium	11.50866, 77.86053	30 KM/Hr
16	Turn	Medium	11.54750, 77.85273	30 KM/Hr
17	Turn	Medium	11.55039, 77.85511	30 KM/Hr
18	Turn	Medium	11.55460, 77.85624	30 KM/Hr
19	Turn	High	11.57893, 77.85430	15 KM/Hr
20	Turn	Medium	11.57903, 77.85441	30 KM/Hr
21	Turn	Medium	11.58022, 77.85744	30 KM/Hr
22	Turn	Medium	11.60727, 77.84032	30 KM/Hr
23	Turn	Medium	11.60972, 77.84164	30 KM/Hr
24	Turn	Medium	11.64064, 77.84611	30 KM/Hr
25	Turn	Medium	11.64535, 77.85122	30 KM/Hr
26	Turn	Medium	11.65473, 77.85327	30 KM/Hr
27	Turn	Medium	11.66467, 77.85599	30 KM/Hr
28	Turn	Medium	11.67301, 77.86024	30 KM/Hr
29	Turn	Medium	11.67326, 77.86058	30 KM/Hr
30	Turn	Medium	11.67373, 77.86230	30 KM/Hr
31	Turn	Medium	11.68171, 77.86676	30 KM/Hr
32	Turn	High	11.68490, 77.86893	15 KM/Hr
33	Turn	High	11.68472, 77.87281	15 KM/Hr
34	Turn	Medium	11.69302, 77.87453	30 KM/Hr
35	Turn	High	11.69458, 77.87589	15 KM/Hr
36	Turn	Medium	11.69475, 77.87583	30 KM/Hr
37	Turn	High	11.69494, 77.87544	15 KM/Hr
38	Turn	High	11.69501, 77.87546	15 KM/Hr
39	Turn	High	11.69553, 77.87440	15 KM/Hr
40	Turn	Medium	11.69915, 77.87478	30 KM/Hr
41	Turn	Medium	11.69941, 77.87503	30 KM/Hr
42	Turn	Medium	11.76013, 77.89028	30 KM/Hr
43	Turn	Medium	11.76020, 77.89070	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
44	Blind Spot	Blind Spot	11.75947, 77.89214	10 KM/Hr
45	Turn	High	11.81889, 77.93336	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Parusuraman Hospital	11.697164, 77.877792	30 km/h	Medium
2	hospital	JP Nature Cure Hospital Siddha and Naturopathy	11.7002949, 77.8770897	30 km/h	Medium
3	hospital	SSS Hospital	11.7501471, 77.8832809	30 km/h	Medium
4	clinic	DSR Clinic	11.7603176, 77.8908853	30 km/h	Medium
5	hospital	Srilakshmi Hospital	11.7601, 77.892272	30 km/h	Medium
6	hospital	Government Hospital	11.758158, 77.894528	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60348, 77.83967



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46303, 77.84945



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.49390, 77.86748



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.49404, 77.86724



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.50866, 77.86053



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54750, 77.85273



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55039, 77.85511



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.55460, 77.85624



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.57903, 77.85441



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.58022, 77.85744



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60727, 77.84032



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.60972, 77.84164



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64064, 77.84611



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64535, 77.85122



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.65473, 77.85327



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.66467, 77.85599



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.67301, 77.86024



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67326, 77.86058



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.67373, 77.86230



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68171, 77.86676



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68490, 77.86893



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.68472, 77.87281



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69302, 77.87453



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69458, 77.87589



Risk Type: Turn

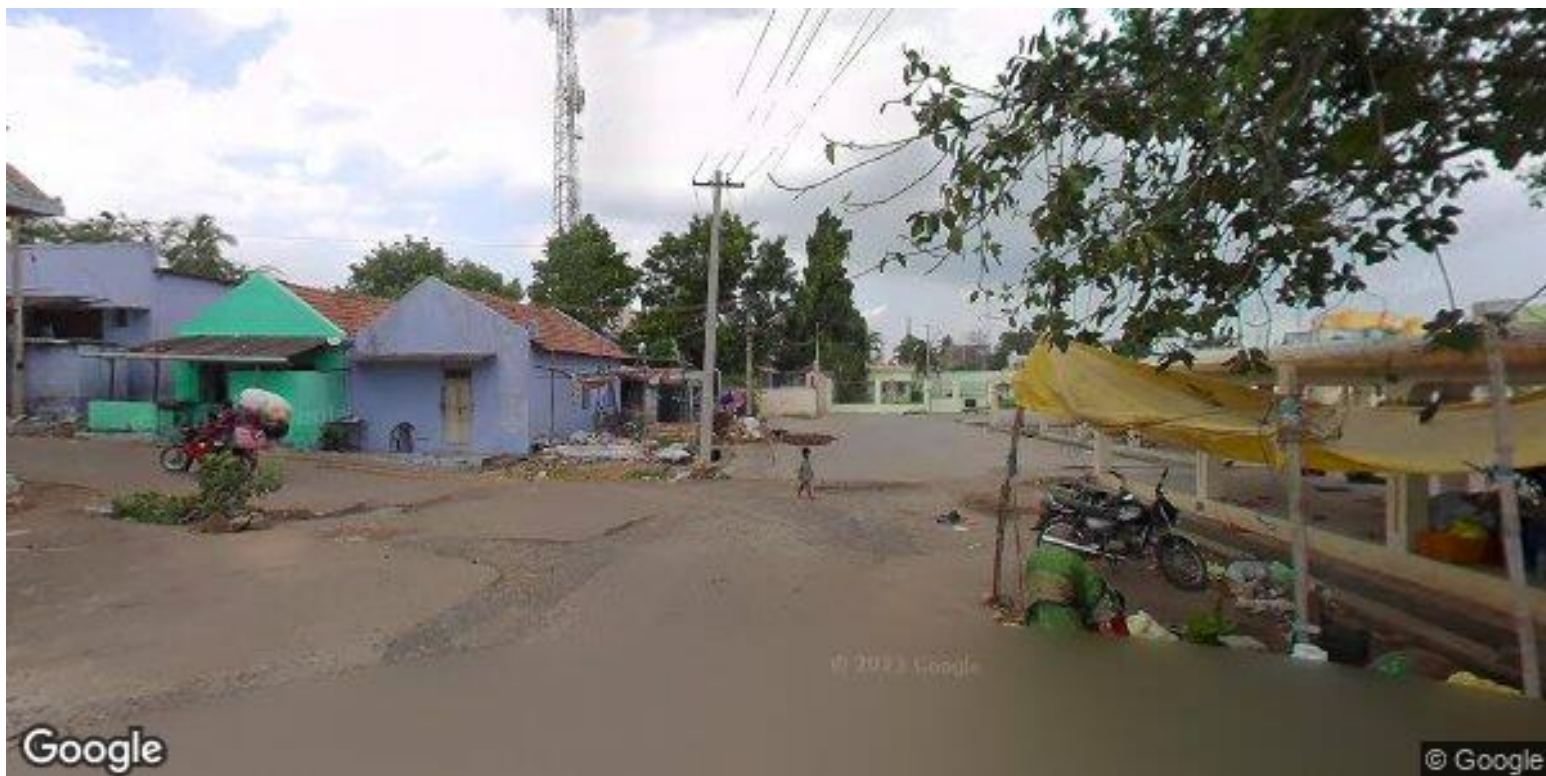
Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69475, 77.87583



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69494, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69501, 77.87546



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.69553, 77.87440



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69915, 77.87478



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.69941, 77.87503



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76013, 77.89028



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76020, 77.89070



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.75947, 77.89214



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.81889, 77.93336
